

NALD Calculations flow chart

$$G' = G_A - 3\Omega \int \frac{\Gamma_{\alpha\beta}(\omega)g(\omega)(\omega^2 - \Omega^2)}{m^2(\omega^2 - \Omega^2)^2 + \nu^2\Omega^2} dw$$

$$G'' = 3\Omega \int \frac{\Gamma_{\alpha\beta}(\omega)g(\omega)\nu\Omega}{m^2(\omega^2 - \Omega^2)^2 + \nu^2\Omega^2} dw$$

Input:

1. $g(\omega) \rightarrow$ density of State
2. $\Gamma_{\alpha\beta}(\omega) = \sum_i |\Xi_i \cdot e_i(\omega)|^2 \rightarrow$ affine force field Correlator
3. $\nu \rightarrow$ friction kernel

Flow chart of NALD Calculations

MD trajectories

- Hessian matrix: $H_{ij}^{\alpha\beta} = \frac{\partial^2 U}{\partial R_i^\alpha \partial R_j^\beta}$ ($U \rightarrow$ potential energy)
After diagonalization of the Hessian, we obtain
- 1. Eigen value (ω_i)
2. Eigen vector (e_i)

1. DoS Calculation Path: (Code for DoS)

- $g(\omega) = \frac{1}{3N-3} \sum \delta(\omega - \omega_i)$

2. Affine force field correlator: (Code for $\Gamma_{\alpha\beta}$)

- Applied strain $\eta \approx 10^{-6}$

$$\Xi_{i,xy} = \frac{\partial f_i}{\partial \eta_{xy}} \Big|_{\eta \rightarrow 0}$$

- Projection of eigen vector onto affine force field

$$\Xi_{k,xy} = e_k \cdot \Xi_{k,xy}$$

- Affine force field Correlator

$$\Gamma_{\alpha\beta}(\omega) = \langle \Xi_{k,xy}^2 \rangle$$

Final expression for the modulus obtained by performing the integration using the input:

$$g(\omega), \quad \Gamma_{\alpha\beta}(\omega), \quad \nu \quad \rightarrow \quad G', \quad G''$$

(Code for G' and G'' (Integration))